

## Module 6 — Lagrange Points & Complex Orbits (Orbit Academy v5.5)

*Printable edition. The interactive version — live simulator, self-checking questions — is the course website; open the matching tut/worksheet pages there. Dynamic diagrams appear here as bracketed figure notes.*

### What you will learn

- why an object at a **Lagrange point** keeps the **same 27.3-day period as the Moon**, even though it sits at a different distance from Earth.
- how, along the Earth-Moon line, the Moon's pull **cancels part of** Earth's at L1 but **adds to** it at L2 — so both still match the lunar period.
- why **L1 and L2 are unstable** (a golf ball on a basketball) so spacecraft fly looping **halo orbits** and nudge often, while **L4/L5** are stable.
- what **orbital resonance** is, using **TESS** — a 2:1 lock with the Moon that keeps it stable for years.
- why the tangled, time-changing **gravity of Earth + Moon + Sun** makes many trajectories **chaotic** — tiny differences blow up — and why that makes cislunar tracking hard.

### Quick-start tutorial

Until now you have orbited **either** the Earth **or** the Moon. Out in cislunar space you orbit **both at once**. What steers a spacecraft there isn't a single tidy pull — it's a **complex, ever-changing gravitational landscape** made by the Earth and the Moon together (and, more faintly, the Sun, and fainter still the other planets and big asteroids).

Deep inside the Earth's or the Moon's Hill sphere, that body is overwhelmingly in charge and everything else is a small nudge — a **perturbation**. But between them, near the boundary, the two pulls are comparable and the motion gets genuinely intricate. On top of all this, neither body is a perfect sphere: their real shapes and lumpy interiors put tiny extra crinkles in the landscape.

### Lagrange points: matching the Moon's pace

In this landscape there are five special spots — the **Lagrange points**, named for an 18th-century French mathematician — where the combined pull of Earth and Moon lets an object circle the Earth in **exactly one lunar month (27.3 days)**, the same as the Moon. Because they keep pace with the Moon, they sit at fixed places on the rotating Earth-Moon line. (There's an analogous set of **Earth-Sun** Lagrange points, where the period matches Earth's **one-year** trip around the Sun.)

**Picture this:** Picture a **bead on a rigid rod** that sweeps around once a month, with Earth at one end and the Moon out along it. The bead is forced to stay on the rod. Slide it to just the right spot and the leftover gravity — Earth's pull minus or plus the Moon's — is exactly what's needed to swing it around with the rod. That balance spot is a Lagrange point.

How can points at *different* distances all take 27.3 days? A moonless object closer in (like L1) would normally orbit **faster** than a month; one farther out (like L2) would orbit **slower**. The Moon fixes the mismatch. At **L1** (between Earth and Moon) the Moon pulls *outward*, partly cancelling Earth, so the pull is gentler and the object slows to a monthly pace. At **L2** (beyond the Moon) both bodies pull the same way and *add*, so the stronger pull speeds a would-be-slow object up to a monthly pace. The tool’s force-balance view shows this with arrows.

[Figure: Along the Earth–Moon line: at L1 the Moon’s pull points back toward the Moon (outward from Earth), opposing Earth; at L2 both pulls point inward and add. Either way the leftover pull matches a 27.3-day orbit.]

**Stable and unstable balance points**

**L4 and L5** — sixty degrees ahead of and behind the Moon — are **stable**: nudge an object and it drifts back rather than away. The reason is easiest to see if you stop thinking about balance points and just watch the object move. An object at L4 is **orbiting Earth** on an ordinary 27.3-day path, the same period as the Moon’s, sixty degrees ahead of it. Nudge it and it now has a slightly different orbit — so it slides forward or back along the Moon’s orbit, but the Moon’s pull tugs it one way and Earth’s the other, and the two together bend it into a long slow loop that returns. Nothing is perched anywhere; it is a satellite whose orbit keeps getting bent back. That is why dust and debris can collect there on their own. **L1, L2, and L3** are **unstable** — balancing there is like a **golf ball perched on a basketball**. A real mission (like the CAPSTONE spacecraft near the Moon) can’t just park; it flies a slow looping **halo orbit** around the empty point and taps its thrusters now and then to stay. A bonus: a spacecraft this close to the Moon shows the **same illuminated phase the Moon does** at that moment — same Sun angle, so a full moon means a “full” spacecraft.

**Complex orbits, resonance, and chaos**

Not everything sits at a balance point. **TESS**, a NASA telescope, rides a stretched orbit locked in a **2:1 resonance**: it laps the Earth twice for every once the Moon goes around, always meeting the Moon in the same relative spots so the Moon’s tugs cancel out over time instead of building up. That careful bookkeeping keeps it stable for years. But stray from such special paths and the Earth+Moon field turns **chaotic**: two objects released almost identically drift wildly apart within days. You’ll watch that happen — it’s the deep reason keeping track of things in cislunar space is so hard. (You’ll also recognize the Moon’s **tidal lock** here — the same face always toward Earth — as one more consequence of these gravitational tugs.)

**The five Earth-Moon Lagrange points**

undefined

| Point     | Where it is                                                  | What’s special                                                                                    |
|-----------|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>L1</b> | Between Earth and Moon, ~84.9% of the way out (~326,000 km). | Moon’s pull opposes Earth’s → gentler net pull → matches the 27.3-day period.<br><b>Unstable.</b> |

| Point          | Where it is                                                              | What's special                                                                                                   |
|----------------|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>L2</b>      | Beyond the Moon, ~116.8% of the Earth-Moon distance (~449,000 km).       | Earth + Moon pulls add → stronger net pull → matches 27.3 days. <b>Unstable</b> (halo orbits + station-keeping). |
| <b>L3</b>      | On the far side of Earth, opposite the Moon.                             | Matches the lunar period on the other side. <b>Unstable</b> ; little practical use.                              |
| <b>L4 / L5</b> | 60° ahead of and behind the Moon along its orbit (equilateral triangle). | Also match 27.3 days — and are <b>stable</b> , so dust and debris can collect there.                             |

## PART A • What makes cislunar space complex

*You now orbit Earth and Moon together, in an ever-changing gravity landscape.*

### Exercise 1 — One landscape, two attractors

In Module 5 you saw the tug-of-war: near the Earth, Earth wins; near the Moon, the Moon wins. Now zoom into the **space between**, where their pulls are comparable. Here there's no single "center" to orbit — a spacecraft moves through a **combined gravity landscape** shaped by both bodies at once, and that landscape **turns** as the Moon goes around.

The Sun adds a gentler influence, and even the other planets and big asteroids nudge things at a tiny level. And because Earth and the Moon aren't perfect spheres, their real shapes wrinkle the landscape a little more. The upshot: cislunar trajectories can be far richer than the clean ellipses of the earlier modules.

**Predict first:** Before you look: out between Earth and the Moon, do you expect an object's path to be a simple repeating ellipse like the ones you built earlier — or something more complicated?

#### Try it in the simulator:

Open the tool on the **five Lagrange points** scenario. Switch to the **Earth-Moon rotating (synodic)** frame and speed time up, then flip to **ECI** and back to feel how the whole landscape turns with the Moon.

**What to look for:** in the rotating frame the Earth, Moon, and the five marked points all hold still; in ECI the Moon (and the points) sweep around once a month.

#### While you watch, think about:

- Which single frame makes this system easiest to reason about, and why?
- What's the biggest influence on a spacecraft sitting halfway to the Moon — and what are the smaller "perturbations"?
- Why does it help that the Lagrange points hold still in the rotating frame?

**Check your understanding:** Out in cislunar space (between Earth and Moon), what determines a spacecraft's trajectory?

- Only Earth's gravity, since the Moon is far too light to matter out there
- The combined, shifting pull of Earth and Moon together, plus smaller solar nudges

- Only the Moon’s gravity, once you pass the halfway point on the way out
- Essentially no gravity at all, so the craft coasts in a straight line

## PART B · The five Lagrange points

*Five spots that keep the Moon’s pace — and why.*

### Exercise 2 — Five points that keep the Moon’s pace

Scattered through this landscape are five special spots — the **Lagrange points** — where the combined pull of Earth and Moon lets an object circle Earth in **exactly one lunar month**. Because they keep pace with the Moon, they’re fixed on the rotating Earth-Moon line: in the synodic frame they never move.

Three of them (**L1, L2, L3**) sit right on the Earth-Moon line; two (**L4, L5**) sit 60° ahead of and behind the Moon, forming equal-sided triangles with Earth and Moon. The definitions table above lists all five.

**Predict first:** A point closer to Earth than the Moon would, on its own, want to orbit **FASTER** than a month; a point farther out would want to orbit **SLOWER**. How do you think all five points can still share the Moon’s 27.3-day period?

#### Try it in the simulator:

In the **five Lagrange points** scenario, switch to the synodic frame. Identify L1, L2, L3 on the line and L4/L5 off to the sides. Speed up time: the ▲ markers hold their geometry fixed as the whole system turns underneath — which the frozen frame makes look perfectly still. Flip to **ECI** for a moment to see the same pattern sweep around once a month.

**What to look for:** the five ▲ markers keep their pattern locked to the rotating Earth-Moon line — an object placed at any of them would circle Earth at the Moon’s pace, holding station on that line.

#### While you watch, think about:

- Which points are on the Earth-Moon line, and which are off it?
- L4 and L5 make equal-sided triangles with Earth and the Moon — why might that symmetry matter?
- What would it mean, physically, for an object to “hold still” in this rotating frame?

**Check your understanding:** What do all five Lagrange points have in common?

- They all have zero gravity, which is what makes them balance points
- An object there orbits Earth with the Moon’s own 27.3-day period, holding station
- They all sit the same distance from Earth, spaced evenly around the orbit
- They are the only places a spacecraft can survive the radiation environment

### Exercise 3 — The force balance at L1 and L2

Here’s the crux — and it answers the puzzle from the last exercise. Consider a **bead threaded on a rigid rod** that sweeps around once a month, Earth at the center and the Moon out along it. The bead is *constrained* to the rod (that’s the picture the “release on

the line” demo quietly assumes). Along that rod, gravity has to supply exactly the pull needed to swing the bead around with the rod.

At **L1**, between Earth and Moon, the Moon lies farther out, so its pull points *outward* — back toward the Moon, opposing Earth. The two partly cancel, leaving a **gentler** net inward pull. That’s perfect for a point this close, which would otherwise be dragged around too fast. At **L2**, beyond the Moon, both bodies pull the same way and **add**, giving a **stronger** pull — perfect for a point this far out, which would otherwise lag behind. So “balance” never means zero pull; it means the *leftover* pull matches a 27.3-day loop.

**Predict first:** At L1, the Moon sits farther from Earth than the object does. Which way do you think the Moon’s pull points — toward Earth (adding to Earth’s pull) or away from Earth (opposing it)?

**Try it in the simulator:**

Open the **Force balance at L1 • Moon • L2** scenario. Compare the arrows: at L1, note the Moon’s arrow pointing back toward the Moon (opposing Earth); at L2, note both arrows pointing the same way.

**What to look for:** at L1 the Moon’s pull opposes Earth’s (net pull is reduced); at L2 the two pulls add (net pull is increased) — yet both yield the same 27.3-day period.

**While you watch, think about:**

- At L1, is the net inward pull bigger or smaller than Earth’s pull alone? What about at L2?
- Why does a point closer in need a *gentler* pull to take a full month, while a point farther out needs a *stronger* one?
- Does “balance point” mean there is no net pull on the object? (Careful!)

**Check your understanding:** At the L1 point (between Earth and the Moon), how does the Moon’s gravity affect the net pull on an object there?

- It adds to Earth’s pull, so the net inward force is stronger than at other radii
- It points back toward the Moon, opposing Earth, so the net inward pull is gentler
- It has no measurable effect, being far weaker than Earth’s pull at that distance
- It cancels Earth’s pull exactly, leaving no net force acting on the spacecraft

## Exercise 4 — Stable vs. unstable balance points

Not all balance points are equal. **L4 and L5** (60° ahead of and behind the Moon) are **stable**: give an object there a small nudge and it doesn’t leave — it loops lazily *around* the point, so dust and debris naturally accumulate there. **L1, L2, and L3** are **unstable** — parking there is like balancing a **golf ball on top of a basketball**: dead center it holds, but the slightest push and it rolls off.

That instability is why you can’t just leave a spacecraft sitting at L1 or L2. It has to fly a slow looping path *around* the empty point — a **halo orbit** — and fire its thrusters occasionally to stay near it. You’ll see one next.

Here is the part that trips people up, and the cure is to stop looking at the map and look at the *motion*. A gravity landscape scores **positions** — it answers “how much energy does it cost to be here?” Stability is not a question about position. It is a question about

**where a moving object goes next**, and you cannot read that off a contour map any more than you can predict a thrown ball from a photograph of the ground.

So follow the object instead. Switch to the **ECI (inertial) frame** and look at what sits at L4: a perfectly ordinary satellite, **orbiting Earth on a 27.3-day path** — the Moon’s own period — sixty degrees ahead of the Moon. Nothing is balanced on anything. It is just in orbit.

Now nudge it. A small push changes its orbit slightly, so its period is no longer an exact match and it begins to drift along the Moon’s orbit — forward if you sped it up, backward if you slowed it down. As it drifts it comes closer to the Moon, whose pull now acts on it noticeably, while Earth still pulls it the other way. Those two pulls together bend the drift around, and the object swings back the way it came. Overshoot, drift back, overshoot again: the result is a long slow loop that never gets away. In the co-rotating view that loop is the **tadpole orbit** you will fly in Part D.

That is the whole mechanism, and it lives entirely in the **trajectory under two real gravitational pulls** — Earth’s and the Moon’s. No third force is needed to explain it, and none is acting.

**Predict first:** If L1 and L2 are unstable like a golf ball on a basketball, what do you think a real spacecraft has to do to “stay” at one of them?

**Try it in the simulator:**

Back in the **five Lagrange points** scenario, switch to the **synodic** frame and turn on the **gravity landscape** checkbox. It draws the co-rotating gravity field as a wireframe surface — **Earth and Moon only**. (The Sun is outside this module’s model. Its differential pull is about 1% of Earth’s at the Moon’s distance, and it only takes over far out near **Earth’s** Hill sphere, ~1.5 million km.)

Find the deep funnels at Earth and the Moon, and the **saddle passes** at L1, L2 and L3. Then do the important step: flip to **ECI** and watch what an object at L4 is actually doing. The landscape scores positions; ECI shows you the orbit. Part D lets you run the experiment for real.

**What to look for:** nudge something near **L4 or L5** and it loops back; nudge it near **L1, L2 or L3** and it leaves — which is why real missions at L1/L2 fly a halo orbit and spend fuel to stay. On the landscape L1 reads clearly as a mountain pass between two basins. For L4/L5, trust the ECI view instead: an ordinary Moon-period orbit 60° ahead, bent back each time it drifts by the combined pulls of Earth and the Moon.

**While you watch, think about:**

- Why might dust or small debris pile up at L4/L5 but not at L1/L2?
- An object sitting at L4 is stable against a small nudge. What actually brings it back?
- What’s the trade-off of using an unstable point like L2 anyway (why bother)?
- How is “balancing a golf ball on a basketball” a good picture of an unstable equilibrium?

**Check your understanding:** Which Lagrange points are STABLE (a nudged object stays nearby, looping around the point)?

- L1 and L2

- L3 only
- L4 and L5
- All five are stable

## PART C · Halo orbits & resonance

*How real missions live at unstable points, and what a resonance buys you.*

### Exercise 5 — A near-rectilinear halo orbit (NRHO)

Since you can't sit still at an unstable point, the trick is to **loop around it**. A **halo orbit** is a three-dimensional loop that circles an empty Lagrange point; with occasional small thruster nudges a spacecraft can ride it for years. The exotic member of the family is the **near-rectilinear halo orbit (NRHO)** — the real path NASA's **CAPSTONE** flew and the one the crewed **Gateway** station will use.

An NRHO is dramatically stretched and nearly **vertical**: the spacecraft **skims low over one lunar pole** (a few thousand km) and then swings **far out over the other pole** (tens of thousands of km), taking about **6.5 days** per loop. That shape is chosen on purpose — it stays in near-constant sunlight and keeps an almost unbroken radio line to Earth, while costing very little fuel to hold.

Notice it is **lopsided about the Moon's orbital plane** — close on one side, far on the other. That may feel wrong: the gravity field *is* mirror-symmetric top-to-bottom, so shouldn't the orbit be too? No. The field's symmetry means halos come in mirror-image **pairs** — a "northern" one and a "southern" one — but each individual orbit picks a side. (It's like a symmetric double-well: the pair of low points is symmetric, yet a ball settles into *one* well.) Gateway will fly a southern NRHO; its mirror twin is equally valid. The orbit *is* still perfectly symmetric — just about the plane through the **Earth-Moon line and the poles**, not the orbital plane.

One neat consequence of being near the Moon: the spacecraft is lit by the Sun from the **same angle the Moon is**, so it shows the **same phase** — near full moon it's a "full" spacecraft, near new moon it's dark. Same Sun, same geometry.

**Predict first:** A spacecraft can't rest at an unstable Lagrange point. Before you watch: what shape of motion do you think lets it stay *near* the point without sitting exactly on it? And do you expect the loop to sit evenly above and below the Moon's orbital plane, or lopsided to one side?

#### Try it in the simulator:

1. Open the **near-rectilinear halo (NRHO)** scenario. Watch a few full loops and pay attention to the **pace** of the spacecraft — where does it move fast, and where does it seem to hang?
2. Use the **arrow keys to rotate the view** (←/→ swing around, ↑/↓ tip over the pole). Find the edge-on angle where you can see the loop is **lopsided about the orbital plane** — close on one side, far on the other — not evenly split.
3. Flip through the four reference frames, then open **■ Compare all 4 frames (2×2)**: the same tall loop looks completely different held still against the stars (ECI), against the Earth-Moon line (synodic), and centered on the Moon (MCI).

**What to look for:** the spacecraft never sits on the point — it rides a tall, lopsided loop; it **whips through the close pole pass in hours but lingers for days out at the far end**, so it is actually far from the Moon almost the whole time; rotating the view shows the loop is one-sided about the orbital plane, and the 2×2 view shows how differently it reads in each frame.

**While you watch, think about:**

- Why loop *around* the point instead of sitting on it?
- Here's the counterintuitive part: this is called a **lunar** orbit, yet where does the spacecraft spend **most of its time** — right over the Moon, or far away from it? Why? (Recall Molniya from Module 3: fast at the close point, slow at the far point.)
- The loop isn't mirror-symmetric about the orbital plane — yet the gravity field is. How can a symmetric field allow a one-sided orbit? (Think of the pair: is there a mirror-image halo on the other side?)
- Which of the four frames made the halo's shape easiest to understand, and why?

**Check your understanding:** Why does a spacecraft near an unstable Lagrange point fly a looping “halo orbit” (like the NRHO) instead of parking at the point?

- The point itself is too far out to reach with a practical amount of fuel
- The point is unstable — like a golf ball on a basketball — so it loops around instead
- A halo orbit uses no propellant at all, unlike sitting at the point itself
- The Moon physically blocks the line of sight to the point from Earth

## Exercise 6 — TESS and orbital resonance

Some complex orbits stay stable not by balancing forces but by clever **timing**. NASA's **TESS** telescope rides a stretched, high orbit in a **2:1 resonance** with the Moon: it goes around Earth **twice** for every **once** the Moon does. Because it always returns to the same places relative to the Moon, the Moon's repeated tugs **average out** instead of piling up and wrecking the orbit — so TESS holds its path for years with almost no fuel.

Resonance is a recurring trick in the solar system (it also sculpts the asteroid belt and Saturn's rings). TESS is a clean, modern cislunar example — and a good one to view in different frames, since the same path looks strikingly different depending on what you hold still.

**Predict first:** TESS circles Earth twice for every one lunar orbit, always meeting the Moon in the same spots. Do you think the Moon's repeated tugs will build up and wreck the orbit, or cancel out over time? And before you switch frames: which viewpoint do you think will make the 2:1 pattern most obvious?

**Try it in the simulator:**

1. Open the **TESS** scenario, speed up time, and count laps: the dot rings Earth **twice** for every one lap the Moon makes.
2. Switch to the **Earth-Moon rotating (synodic)** frame and keep watching the dot: every lunar month it meets the Moon with the **same relative geometry** — that repeating rendezvous is the 2:1 lock.
3. Use the **arrow keys to rotate** the view and confirm TESS's orbit is **tilted out of the Moon's plane** (it's an inclined HEO, not co-planar).

**What to look for:** TESS laps Earth exactly twice per lunar orbit, so it returns to the same geometry relative to the Moon each cycle — easiest to catch in the synodic frame, where the Moon holds still while the dot repeats its rendezvous — and rotating shows the orbit is inclined, not in the Moon's plane.

**While you watch, think about:**

- What does “2:1” literally count — orbits of what, versus what?
- Why does always meeting the Moon in the same places keep the orbit stable?
- Why does the synodic frame — where the Moon holds still — make the repeating 2:1 rendezvous easiest to catch?
- How is holding the right frame still (to reveal a repeating pattern) like the resonance itself — a matter of matching rhythms?

**Check your understanding:** TESS is in a 2:1 resonance with the Moon. What does that mean, and why does it help?

- It orbits twice as fast as the Moon, which keeps the two permanently apart
- It laps Earth twice per lunar orbit, meeting the Moon at the same points each time
- It parks exactly at L2 and stays there, held by the resonance with the Moon
- It never comes near the Moon at all, which is what makes the orbit safe

## **PART D · Chaos: the fan release**

*Six release experiments — the L1 knife-edge, stable tadpoles at L4, lunar slingshots, retrograde parking, the free-return figure-8, and Earth's temporary minimoons — and why tracking out here is hard.*

### **Exercise 7 — The knife-edge at L1**

You learned in Part B that L1 is an **unstable** balance point — a ball on a saddle. Now feel what that means. The **L1 knife-edge** release drops **seven objects exactly at L1**, all riding the rotating Earth-Moon line, differing only by tiny velocity kicks — **at most 8 m/s, bicycle pace**. You could not tell them apart by eye.

For days, nothing happens: the balance is real, and they hover together. Then the saddle wins. Which way each object falls is decided entirely by its imperceptible kick: toward the Moon, toward Earth, or holding out longest with no kick at all. Even the **un-kicked** one leaves after ~3 weeks — at an unstable equilibrium, disturbances too small to measure (in the sim, literally the computer's round-off) are eventually enough. Nothing stays at L1 without station-keeping.

**Predict first:** Seven objects at the same point, same speed to within 8 m/s. Will they still be together after two weeks? Where do you think they'll end up?

**Try it in the simulator:**

1. Open the **Fan release** scenario, pick **L1 knife-edge**, and press **► release**. Watch in the **synodic (co-rotating)** frame, where L1 stands still.
2. For the first days they sit as one dot on the **▲ L1** marker. Be patient (or tap . to speed up) — then watch them peel away one by one.

3. Track the fates: some fall to the **Moon** (~day 8-9), some fall **home toward Earth** (~day 11-14), the out-of-plane one lasts to ~day 18, the un-kicked one to ~3 weeks. The “current scatter” readout measures the blow-up live.

**What to look for:** objects that started indistinguishably close hover together for days, then diverge to completely different fates — the saddle amplifies any difference, however small.

**While you watch, think about:**

- How long did the group LOOK stable before the first one left?
- An 8 m/s kick is about riding-a-bike speed. Why does something that small decide between “falls to the Moon” and “falls home”?
- What does this imply for a real spacecraft parked near L1 — can it just sit there?

**Check your understanding:** Seven objects are released exactly at L1 with kicks of at most 8 m/s. What does the knife-edge release show?

- They all stay put at L1 — it is a balance point, so nothing should move
- They hover for days, then tiny kick differences send them to entirely different fates
- They fly apart at once, in directions unrelated to the kicks they were given
- The kicks are far too small to matter, so the group drifts along together

### Exercise 8 — Tadpoles at L4 — the stable pocket

Now run the identical experiment at **L4**, 60° ahead of the Moon — where Part B promised **stability**. The **L4 tadpoles** release uses kicks up to **30 m/s**, nearly four times what doomed the L1 set.

The result is completely different: instead of falling away, the objects **librate** — they trace lazy closed loops around L4, weeks per lap, shaped like tadpoles in the rotating frame. Bigger kick, bigger loop; but they stay. This is why **Trojan asteroids** have camped at Jupiter’s L4/L5 for billions of years, with nobody station-keeping them. One object in the set was kicked **50 m/s** — and it breaks out and wanders off along the Moon’s orbit: the stable pocket is real, but it has a **finite rim**.

**Predict first:** Same experiment, but at L4 with kicks up to 4× bigger. Based on Part B, what do you expect this time?

**Try it in the simulator:**

1. Pick **L4 tadpoles** and press ► **release**. Stay in the **synodic** frame — libration is invisible in any other view.
2. Speed up with . and watch for weeks of sim time: the objects loop around the L4 marker without leaving.
3. Find the **red** object — the 50 m/s one. Watch it leave the tadpole region and drift along the Moon’s orbit (it makes a startlingly close pass by the Moon).

**What to look for:** kicks that would have doomed the L1 set just set the size of a closed libration loop around L4 — only the 50 m/s outlier escapes the pocket.

**While you watch, think about:**

- The L1 set scattered in days; the L4 set is still together after months. What single word explains the difference?

- Where does a “tadpole loop” exist — in the inertial view, or only in the rotating frame? Why?
- If you had to park a satellite in cislunar space for a decade with no fuel, where would you put it?

**Check your understanding:** The same kicks (and bigger) that scattered the L1 release only make objects at L4 loop in place. Why?

- L4 has stronger local gravity, which holds displaced objects down against the kick
- L4 is stable: a displaced object librates in a closed tadpole loop rather than leaving
- The Moon shields objects at L4 from Earth’s pull, removing the destabilising force
- It is a coincidence of this particular release, and other kicks would scatter them

## Exercise 9 — Lunar scatter — slingshots and escape

The knife-edge and tadpole releases never left the Earth-Moon neighborhood. Now launch a fan of objects on a **transfer orbit whose far point just reaches the Moon** — spread slightly in aim, timing and **out of the orbital plane**. They first **coast up one clean arc** for about nine days, then all arrive near the Moon in quick succession. When an object passes **close to the Moon**, the Moon’s gravity swings it hard — a **gravitational slingshot** (the same trick real missions like the Voyagers used to pick up speed). A near-miss can whip an object off in almost any direction; a direct hit ends the trip.

Here’s the striking part: a slingshot can hand an object **so much extra speed** that it is flung far beyond Earth’s **Hill sphere** — the ~1.5-million-km bubble inside which Earth’s gravity, not the Sun’s, is in charge. Past that line, in the real Solar System, the Sun takes over and the object never comes back. In this run the **yellow** object skims about **300 km above the lunar surface** — Apollo-8 altitude — and gets thrown onto a ~4-million-km orbit. Whether that happens is exquisitely sensitive to the exact aim: the **red** object, aimed only slightly differently, instead whips a tight **hairpin** around the far side and stays.

The tool tallies a “**flung away so far**” count: objects that are either formally unbound (orbital energy  $\geq 0$ ) or carried beyond the Hill sphere.

**Predict first:** Seven objects on nearly-identical transfer orbits toward the Moon. Every one makes a close pass. **Predict first:** how many of the seven get **flung away** versus stay looping in the system? Write down your guess.

### Try it in the simulator:

1. Open the **Fan release** scenario and pick **lunar scatter**, then press ► **release**. They **coast up one clean arc together** (~9 days) before reaching the Moon.
2. **Before they arrive, switch to the Moon-centered inertial (MCI) frame** — that puts the camera on the Moon, so each flyby draws its true shape: a clean gravity-bent hyperbolic swing past the Moon. (In Earth-centered views the Moon itself moves during the encounter and the geometry is much harder to read.)
3. Slow down with , as they close in and watch each pass: the **red** hairpin around the far side (~1.4 lunar radii), the **yellow** 300-km skim, and the wider deflections out to ~13 lunar radii. Every turn is the Moon *pulling*, never pushing.
4. Speed back up and let it run for weeks of sim time. Watch the “**flung away so far**” readout — a green banner announces the first object past the Hill sphere.

**What to look for:** all seven make a close lunar pass, then split fates: most stay looping, but the deepest skim is flung past Earth's Hill sphere — and in MCI each swing is a clean pull-around, nothing ever bounces.

**While you watch, think about:**

- How close did your prediction come to the actual tally of flung-away vs. staying?
- What gives an escaping object its extra speed — where does that energy come from? (Hint: the Moon is moving.)
- Why can two objects launched only a few degrees apart end up with completely different fates?
- How is this the same physics a mission uses to get a “free” speed boost from a flyby?

**Check your understanding:** In the lunar-scatter release, one object gets “flung away.” What does that mean, and how did it happen?

- Its tracking data was lost, so the simulation could no longer label the object
- A grazing lunar flyby threw it clear past Earth's Hill sphere, where the Sun takes over
- It exhausted its propellant and drifted away from the group it started with
- The Moon captured it into a lunar orbit, removing it from the Earth-centred set

### **Exercise 10 — See the scatter in 3-D — rotate for the out-of-plane view**

Seen from straight above (the default top-down view), the scattering looks like it happens in a flat plane — but it doesn't. Each object was launched with a slightly **different tilt out of the orbital plane**, so after the slingshot their paths splay out in **all three dimensions**, above and below the Moon's orbit. The flat view hides that entirely.

This is exactly the trap real trackers fall into: a single camera angle (or a single sensor) flattens a 3-D reality into 2-D, and two objects that look like they're in the same place can actually be tens of thousands of kilometres apart **out of the plane**. Rotating your viewpoint is the cheapest way to recover that missing dimension — and it's why cislunar surveillance wants observers at more than one vantage point.

The **arrow keys tilt and spin the camera** (←/→ swing around, ↑/↓ tip up and over the pole). Give the scene a good tumble while the objects are scattering and the out-of-plane structure jumps out.

**Predict first:** The top-down view makes the scatter look flat. Before you rotate: do you expect the seven paths to actually lie in one plane, or to spread **above and below** the Moon's orbit? By roughly how much — hundreds of km, or tens of thousands?

**Try it in the simulator:**

1. Run the **lunar scatter** release again and let the objects reach the Moon and begin scattering.
2. Now hold the **← and → arrow keys** to swing the camera around the side, then **↑ / ↓** to tip the view up over the pole — watch the trails lift off the flat plane into a 3-D spray. (Mouse: shift-drag or right-drag rotates; wheel zooms; R resets the view.)
3. Find the edge-on angle where the orbital plane looks like a thin line — the out-of-plane objects clearly stand off it, above and below.

4. For a second perspective, open the **■ Compare all 4 frames (2×2)** view and rotate one panel independently while the others hold still — the same scatter, read from four viewpoints at once.

**What to look for:** the paths are genuinely three-dimensional: rotating the view reveals objects arcing well above and below the Moon's orbital plane, spread that was completely invisible from straight overhead.

**While you watch, think about:**

- How much did rotating the view change your sense of where the objects actually are?
- If you only ever saw the top-down view, what mistake might you make about two objects that appear to overlap?
- Why would having **two** observers at different angles beat one, for pinning down a cislunar object's true position?
- The launch tilts were tiny — why does a small out-of-plane nudge grow into a large out-of-plane spread after the Moon flyby?

**Check your understanding:** Rotating the camera during the lunar-scatter release shows that the objects' paths are...

- Exactly in one flat plane, since the Moon's small tilt makes no real difference
- Spread through all three dimensions, well above and below the Moon's orbit plane
- Always pointing straight back at Earth, whatever the release direction was
- Identical to one another, merely drawn at different sizes by the perspective

### **Exercise 11 — Parking at the Moon: backwards is better**

Suppose you want to **leave a satellite near the Moon** for years. Around Earth (Modules 1-3) any circular orbit works. Around the Moon it's not so simple — **Earth's tide is always tugging**, and far enough out it can strip a satellite away entirely.

The **DRO vs prograde** release parks seven satellites around the Moon: three **retrograde** (circling the Moon backwards, as seen in the rotating frame) at 30, 40 and 55 thousand km, three **prograde** twins at the same radii, and one close-in prograde at 20,000 km. The retrograde ones are **Distant Retrograde Orbits (DROs)** — famously stable; it's where **Artemis I parked the Orion capsule** in 2022. The prograde twins are doomed: Earth's tide pumps them up and strips them off (one even spirals into the Moon). Only close in, where the Moon's grip is overwhelming, does prograde survive.

**Predict first:** Same radius, same speed — only the direction (prograde vs retrograde) differs. Do you expect that to matter? Which sense survives?

**Try it in the simulator:**

1. Pick **DRO vs prograde** and press ► **release**. Watch in **Moon-centered inertial** or the **synodic** frame.
2. Speed up with .. Within days the far prograde satellites distort, swing wide, and are **stripped** — the "stripped from the Moon" counter tracks them (one is flung 2.3 million km; the 30k prograde one craters into the Moon around day 39).
3. Let it run for weeks: every **retrograde** orbit just keeps looping, only gently flexed by the tide. So does the close-in 20k prograde one.

**What to look for:** direction is destiny: at the same radii, retrograde (DRO) orbits persist for months while prograde ones are stripped away or crash — and close in, both survive.

**While you watch, think about:**

- Same radius, same speed, opposite direction — why does retrograde resist the tide so much better? (Think about how fast the satellite sweeps past the Earth-Moon line in the rotating frame.)
- Why did NASA choose a DRO to park Orion during Artemis I?
- The 20,000-km prograde orbit survived. What does that tell you about WHERE the Earth's tide matters?

**Check your understanding:** Two satellites orbit the Moon at 40,000 km — identical except one is prograde, one retrograde (a DRO). What happens over the following weeks?

- Both are equally stable, because orbital direction cannot affect stability
- The prograde one is stripped away within days; the retrograde DRO persists
- The retrograde one fails immediately, since orbiting backwards is unstable
- Both are stripped away, because nothing can orbit the Moon that far out

### Exercise 12 — The free-return figure-8 — 6 m/s from home

One more release, and it's the most famous trajectory in spaceflight. A **free-return** path leaves Earth, loops around the **far side of the Moon**, and falls home to re-entry — **with no burn at the Moon**. Gravity does everything after departure. Apollo rode free-returns because they're fail-safe: if the engine dies (Apollo 13!), you still come home.

The  $\infty$  **free return** release sends seven ships off a 330-km parking orbit with departure speeds spanning just **~6 m/s in total** — the pace of a brisk walk, on top of 10,800 m/s. That whisker is the whole difference between re-entering at home, missing Earth by 80,000 km, being stranded near lunar distance, or hitting the Moon.

**Predict first:** Seven ships, departure speeds within 6 m/s of each other. How different can their fates be?

**Try it in the simulator:**

1. Pick  $\infty$  **free return** and press ► **release**. Watch in the **Earth-centered inertial** frame to see the figure-8 shape.
2. Follow them out (~5 days to the Moon). **Green** and **cyan** thread the figure-8 and **re-enter around day 8** — a green splashdown banner marks the first one home. Keep watching: **pale** straggles home around day 13 and **pink** around day 18 — **4 of 7 eventually make it back**, but only the true free-returns arrive on a schedule a crew could survive on.
3. Track the failures: **red** cuts inside one lunar radius and craters (~day 5); **yellow** and **purple** are slung wide and still haven't come home after *months*. The "came home so far" counter keeps score.

**What to look for:** a ~6 m/s spread at departure fans out into on-time re-entry, days-late re-entry, months of stranding, and a lunar impact — and the ships that do come home used no propellant at all after leaving Earth.

**While you watch, think about:**

- Why is a free-return trajectory the safest way to send people to the Moon?

- The speed differences are  $\sim 0.05\%$  of the departure speed. Why does the Moon amplify them so violently? (Which earlier release does this echo?)
- Apollo 13's explosion happened on the way out. Why did the crew survive?

**Check your understanding:** What makes a free-return trajectory special — and why did Apollo use one?

- It is the fastest available route to the Moon, which is why Apollo chose it
- After the departure burn gravity alone loops the ship round the Moon and home
- It uses the Moon to fling the ship right out of the Earth-Moon system
- It needs continuous thrust to hold the figure-8 shape all the way round

### Exercise 13 — Earth's temporary minimoons

Earth occasionally has **more than one moon**. Small asteroids drifting past can be **ballistically captured**: a well-timed lunar flyby steals enough of their energy to leave them gravitationally bound to Earth — a **minimoon** — until a later encounter hands the energy back and releases them. This really happens: asteroid **2006 RH120** orbited Earth for about 11 months in 2006-07, and **2020 CD3** stayed for roughly two years. No engines, no burns — pure gravity.

The **minimoons** release sends seven objects sunward-in from a million km out. Every one is slightly **hyperbolic** — NOT bound to Earth — with identical speed and identical aim. The only thing that differs is **which day each arrives**, which changes where the Moon happens to be. Watch the “captured right now” counter: it tracks how many currently have negative orbital energy — i.e., how many moons Earth has at the moment.

**Predict first:** Seven identical unbound arrivals, differing only in arrival day. Can gravity alone make any of them orbit Earth? For how long?

**Try it in the simulator:**

1. Pick **minimoons** and press ► **release**. Watch in the **Earth-centered inertial** frame, zoomed well out.
2. A green banner marks the first **ballistic capture**. Watch the counter rise and fall as objects are caught and released.
3. Be patient: the infall takes  $\sim 3$  weeks of sim time ( $\sim 25$  s at the default speed) before anything can be captured — a **green halo ring** lights around each object while it is bound to Earth. Follow individual fates: **yellow** and **pink** stay captured for over a year of sim time; **green** and **purple** are minimoons for  $\sim 3$  weeks; **cyan** sails straight through; **red** arrives at exactly the wrong moment and hits the Moon.
4. Note how irregular the captured orbits look — huge, lopsided, never repeating. Real minimoons are like this, which is one reason they're hard to spot.

**What to look for:** objects that arrived unbound end up looping Earth for weeks to over a year — captured and eventually released by nothing but lunar flybys — while identical twins arriving on other days pass straight through.

**While you watch, think about:**

- Where did a captured object's excess energy go? (Something else must have gained it.)
- All seven were identical except arrival day. What does that tell you about predicting whether a passing asteroid will be captured?

- Why are real minimoons so hard to discover? (Think size, brightness, and their weird orbits.)

**Check your understanding:** How does a passing asteroid become a temporary “mini-moon” of Earth, with no propulsion involved?

- It is slowed by friction with the thin interplanetary medium near Earth
- A well-timed lunar flyby hands energy to the Moon, dropping it below escape
- Earth’s magnetic field grips the asteroid and holds it in a temporary orbit
- It cannot happen at all — capture always requires a propulsive burn

## Exercise 14 — Why cislunar tracking is hard

Put the pieces together. In simple Earth orbit, a handful of numbers (the elements from Module 3) predict where something will be far into the future. In the tangled, chaotic cislunar field, that breaks down: a small uncertainty in where an object **is now** becomes a **huge** uncertainty in where it will be later — exactly the fan-release behavior.

That’s why keeping **custody** of objects in cislunar space (a theme from Module 4) is so demanding: you must observe them **often**, because predictions go stale fast. It’s a frontier problem for space domain awareness, and it’s the practical payoff of this whole module.

**Predict first:** Given what the fan release showed, do you expect it to be easier or harder to predict a cislunar object’s future position than a simple Earth-orbiting satellite’s — and why?

### Try it in the simulator:

Reflect on the fan release you just ran. Imagine each dot is a real object you’re trying to track, and you only get to measure its position occasionally.

**What to look for:** because paths diverge so fast, a position measured today tells you less and less about tomorrow — you have to keep re-observing.

### While you watch, think about:

- How does chaos turn a small “where is it now?” error into a big “where will it be?” error?
- Why does that force you to observe cislunar objects more often than Earth-orbit satellites?
- How does this connect to “custody” from Module 4?

**Check your understanding:** Why is predicting a cislunar object’s future position so much harder than for a simple Earth-orbiting satellite?

- Cislunar space has no gravity worth modelling, so the motion is unpredictable
- The chaotic two-body field amplifies small position errors into huge future ones
- Cislunar objects travel so fast that observations are always out of date
- There is no practical way to observe objects at those distances at all

## PART E • How to orbit around nothing

*The libration zoo: non-Keplerian orbits around an empty point, set by a single number. Maximum weirdness.*

## Exercise 15 — How to orbit around nothing

One last, mind-bending idea to end on. Every orbit you’ve met until now goes **around a body** — a satellite around Earth, the Moon around Earth, a probe around the Moon. A **halo orbit goes around nothing at all**. It circles a **Lagrange point** — an empty spot in space with no mass in it. There is no body at the focus because there is **no focus**: the spacecraft is simply balanced in the combined pull of Earth and Moon, tracing a loop around a point where, if it stopped, it would feel almost no net force. (The showcase NRHO you just watched visibly wraps the Moon — no contradiction: the family shades continuously from Moon-hugging NRHOs at one end to halos looping a truly empty point at the other.)

That means these are **not Keplerian orbits**. Kepler’s ellipses — and the whole six-element / **TLE** description from Module 3 — assume one dominant central body. Here there are two, and the field is time-changing, so **none of that machinery applies**: no semi-major axis, no eccentricity, no fixed ellipse, no TLE. Astonishingly, the entire family of halos around a Lagrange point needs just **one number** to pin it down — the **energy** (the Jacobi constant). Set that single knob and the size, the shape, how far it climbs out of the plane, and the period are **all determined**. One number, versus the six-plus of a normal orbit. (The tool — a simplified, linearized model, as its own panel admits — hands you **two amplitude sliders and a frequency ratio** instead; but pick a whole-number ratio like the halo’s 1:1 and it locks the two sliders together, leaving exactly one free knob: the one-parameter family in disguise.)

And notice something as it circles: in the rotating frame the loop holds a fixed orientation, so in the real (inertial) sky its tilt **swings around once a month**, keeping pace with the Earth-Moon line. That turning takes a real **torque** — supplied by the **Moon’s off-axis gravity** (the Moon doesn’t pull straight through Earth, so its tug twists the orbit around). Same idea as the bead on the rotating rod: real gravity, doing the turning.

**Predict first:** A halo orbit circles a Lagrange point — a spot with no object in it. Before you explore: how many numbers do you think it takes to describe the whole family of these orbits — the usual six orbital elements, or something very different?

### Try it in the simulator:

1. Open the **Libration zoo — the L1 orbit family** scenario. Drag the **↔ in-plane amplitude ( $A_x$ )** slider and watch the loop grow smoothly — and notice the note: on a whole-number ratio the two sliders are **locked together**, so dragging either walks you along a one-parameter family. That linked pair is the single knob in disguise.
2. Make sure the frequency ratio is on **1:1 (halo)** and raise the **↑ out-of-plane ( $A_z$ )** slider: the in-plane and out-of-plane rhythms match, so the path closes into a clean, tilted 3-D halo that climbs far out of the plane. **Rotate with the arrow keys** to see the 3-D shape.
3. Now click **irrational** with the **long trail** checkbox on: the two rhythms never line up, so the loop **never closes** — it endlessly fills a 3-D band. Confirm the orbit is **not centered on any body**; it wraps the empty **▲ L1 point** (use the **▲ Lagrange points** overlay checkbox to see the point it circles).
4. Watch one full lap in the **ECI** frame, then the **synodic** frame: the loop holds still against the Earth-Moon line but sweeps around once a month against the stars — the precession the Moon’s torque produces.

**What to look for:** the whole family sweeps out from the locked amplitude sliders — at a whole-number ratio one setting fixes everything, echoing the true one-parameter (energy) family; the orbit circles an empty point (no body at the center), never closes unless the two rhythms lock, and its orientation turns once a month in the inertial sky.

**While you watch, think about:**

- Why can't a TLE (the six-element set from Module 3) describe a halo orbit? What assumption behind Kepler's ellipse breaks here?
- A normal orbit needs six-plus numbers; this family needs one. What happened to the other degrees of freedom? (Hint: the rotating two-body field fixes the orientation and links size to shape.)
- The loop's orientation turns once a month in the inertial frame, so its angular momentum changes — what supplies the torque? (Careful: it is real gravity, not the rotating frame.)
- In what sense is "orbiting nothing" still just Newtonian gravity — no new physics, only two pulls instead of one?

**Check your understanding:** What is fundamentally different about a halo orbit compared with an ordinary satellite orbit?

- It relies on a kind of gravity that does not exist closer to Earth
- It circles an empty point with no body at the centre, so no TLE can describe it
- It is an ordinary ellipse that only looks strange in a rotating frame
- It needs more orbital elements than a normal orbit rather than fewer

## Key points

- Cislunar trajectories are set by the **combined, time-changing gravity of Earth + Moon** (with the Sun and others as smaller nudges, and even the bodies' non-spherical shapes adding crinkles).
- The five **Lagrange points** are spots where Earth + Moon together give an object the **Moon's own 27.3-day period**, so it holds station on the rotating Earth-Moon line.
- At **L1** the Moon's pull opposes Earth's (gentler net pull); at **L2** the two pulls add (stronger net pull). Either way the leftover pull is exactly what a 27.3-day loop needs — so "balance" does **not** mean zero pull.
- **L1/L2/L3 are unstable** (a golf ball on a basketball) — missions fly small looping **halo orbits** and nudge often. **L4/L5 are stable**, so dust and debris can gather there.
- There are analogous **Earth-Sun** Lagrange points with a **one-year** period.
- A spacecraft near the Moon shows the **same illuminated phase** as the Moon at that moment (same Sun angle).
- The Moon is **tidally locked** — one face always toward Earth — because gravity locked its spin to its orbit. So the Sun sweeps around it once a (synodic) month — that moving sunlight is what drives the phases.
- **Orbital resonance** (e.g. TESS's 2:1 lock with the Moon) can make a complex orbit **stable** by arranging the Moon's repeated tugs to cancel over time.
- Away from such special paths, Earth+Moon gravity is **chaotic**: near-identical starts diverge within days. That sensitivity is why long-term cislunar tracking is genuinely hard.

- A close lunar pass acts as a **gravitational slingshot**: it can fling an object in almost any direction — even far beyond Earth's **Hill sphere** (~1.5 million km), where the Sun's gravity takes over and it never returns.
- **Stability is real estate**: L4/L5 and distant retrograde orbits (DROs) hold objects for months to years with no fuel; L1/L2 shed everything without station-keeping; and a **free-return figure-8** comes home from the Moon on gravity alone.

## Final quiz

*Answers are checked in the online version.*

**Q1.** What is a Lagrange point?

- A place in space where the two gravitational pulls leave no force at all
- A spot where Earth and Moon together give an object the Moon's own 27.3-day period
- The point along a transfer where a rocket exhausts its usable propellant
- The geometric centre of the Moon, about which the whole system balances

**Q2.** At L1 the Moon's pull opposes Earth's; at L2 the two pulls add. Yet both points share the 27.3-day lunar period. Why does that work?

- They do not actually match — the two periods differ by several days
- L1 sits closer in and needs a gentler pull, so opposing pulls help; L2 needs a firmer one
- The Moon has no measurable effect at either point, so both simply follow Earth
- Both points lie at the same distance from Earth, so the periods must agree

**Q3.** Which statement about stability is correct?

- All five Lagrange points are stable, which is why debris collects at each
- L1 and L2 are the stable pair; L4 and L5 shed anything placed near them
- L4 and L5 are stable; L1, L2 and L3 are not — a golf ball on a basketball
- None of the five is stable; every one needs continuous station-keeping

**Q4.** L3 sits on the far side of Earth, directly opposite the Moon. Is it at a smaller or larger orbital radius than the Moon?

- Exactly the same radius as the Moon, being its mirror image about Earth
- A slightly larger radius — it sits a little outside the Moon's own orbit
- A much smaller radius, well inside the Moon's orbit and closer to Earth
- A radius that keeps changing, so the question has no single answer

**Q5.** A near-rectilinear halo orbit (NRHO) skims a few thousand km over one lunar pole and swings ~70,000 km out over the other. Where does the spacecraft spend MOST of its time?

- Right over the pole, close to the Moon — that proximity is the whole point
- Far out at the distant end — the close pass takes hours, the far end days
- Half near and half far, since the orbit is symmetric in time about the Moon
- Evenly spread around the loop, because it travels at a constant speed

**Q6.** The NRHO in the tool is clearly lopsided about the Moon's orbital plane — close on one side, far on the other. Is that a mistake, given that the Earth+Moon gravity field is mirror-symmetric top-to-bottom?

- Yes — a symmetric field must produce a symmetric orbit, so the render is wrong
- No — a symmetric field allows a matched pair, northern and southern, each lopsided
- No — the gravitational field is not actually symmetric from top to bottom
- Yes — real halo orbits are perfectly symmetric about the orbital plane

**Q7.** A halo orbit is NOT a Keplerian ellipse. Which statement best captures why, and what it takes to describe one?

- It is an ordinary ellipse, and six TLE elements describe it perfectly well
- It circles an empty point in the combined field — no central body, so no TLE
- It obeys Kepler’s laws, but about the Moon rather than about the Earth
- It needs more than six elements, because two separate bodies pull on it

**Q8.** A halo loop holds a fixed orientation in the rotating frame, so in the inertial sky it swings around once a month — its angular momentum changes. What supplies the torque?

- No torque is needed at all; the apparent turn is just a change of viewpoint
- The Moon’s off-axis pull, which does not point straight through Earth
- A small onboard thruster, firing continuously to hold the orientation
- Earth’s gravity, which acts more strongly on the near side of the loop

**Q9.** In the lunar-scatter release, an object gets “flung away.” What happened to it?

- It lost radio contact, so the simulation could not follow it any further
- A grazing lunar flyby slung it beyond Earth’s Hill sphere, into the Sun’s domain
- It was captured into a stable lunar orbit and left the Earth-centred group
- It slowed, fell back toward Earth, and re-entered the atmosphere

**Q10.** Artemis I parked the Orion capsule in a Distant Retrograde Orbit (DRO). Based on the DRO-vs-prograde release, why?

- DROs sit closer to the Moon, so they cost less propellant to reach
- Far from the Moon, retrograde orbits resist Earth’s tidal pull and stay stable
- Retrograde orbits move faster, so Orion could begin its return sooner
- Prograde orbits at that distance are prohibited by international agreement

**Q11.** A free-return trajectory (Apollo’s figure-8) is prized because...

- It reaches the Moon using less propellant than any other available route
- After the departure burn, gravity alone carries the ship round the Moon and home
- It lets the ship hold position over one chosen spot on the lunar surface
- It avoids the Moon’s gravity entirely, which removes the navigation risk

**Q12.** TESS stays on its complex orbit for years because it is...

- Parked exactly at L1, where the two pulls hold it in place indefinitely
- In a 2:1 resonance, so the Moon’s repeated tugs average out instead of adding
- Far enough out that no gravitational perturbation reaches it any more
- Firing its thrusters continuously to hold the orbit against perturbations

**Q13.** You release two nearly-identical objects in cislunar space. What does the fan-release demo show, and why does it matter?

- They stay together, which makes tracking either one straightforward
- They diverge quickly — chaos turns small position errors into huge future ones

- They both settle into lunar orbit and remain there indefinitely
- They collide with one another almost immediately after release

## Learn more

- Lagrange point — Wikipedia  
[https://en.wikipedia.org/wiki/Lagrange\\_point](https://en.wikipedia.org/wiki/Lagrange_point)
- Halo orbit — Wikipedia  
[https://en.wikipedia.org/wiki/Halo\\_orbit](https://en.wikipedia.org/wiki/Halo_orbit)
- CAPSTONE (near-rectilinear halo orbit)  
[https://en.wikipedia.org/wiki/CAPSTONE\\_\(spacecraft\)](https://en.wikipedia.org/wiki/CAPSTONE_(spacecraft))
- Lunar Gateway — the southern NRHO it will fly — halos come in mirror-image north-  
ern/southern families; Gateway uses a southern one  
[https://en.wikipedia.org/wiki/Lunar\\_Gateway](https://en.wikipedia.org/wiki/Lunar_Gateway)
- TESS (Transiting Exoplanet Survey Satellite) — 2:1 lunar resonance  
[https://en.wikipedia.org/wiki/Transiting\\_Exoplanet\\_Survey\\_Satellite](https://en.wikipedia.org/wiki/Transiting_Exoplanet_Survey_Satellite)
- Orbital resonance — Wikipedia  
[https://en.wikipedia.org/wiki/Orbital\\_resonance](https://en.wikipedia.org/wiki/Orbital_resonance)
- Tidal locking (why one face) — Wikipedia  
[https://en.wikipedia.org/wiki/Tidal\\_locking](https://en.wikipedia.org/wiki/Tidal_locking)
- Three-body problem & chaos — Wikipedia  
[https://en.wikipedia.org/wiki/Three-body\\_problem](https://en.wikipedia.org/wiki/Three-body_problem)
- Jacobi constant (the one parameter of the family) — Wikipedia — the conserved  
rotating-frame energy that labels each libration orbit  
[https://en.wikipedia.org/wiki/Jacobi\\_integral](https://en.wikipedia.org/wiki/Jacobi_integral)
- Lissajous orbit — Wikipedia — the never-closing quasi-periodic cousins of halo orbits  
[https://en.wikipedia.org/wiki/Lissajous\\_orbit](https://en.wikipedia.org/wiki/Lissajous_orbit)
- Lyapunov orbit (the planar base of the family) — Wikipedia  
[https://en.wikipedia.org/wiki/Lyapunov\\_orbit](https://en.wikipedia.org/wiki/Lyapunov_orbit)
- Why halos are non-Keplerian: the CR3BP — Wikipedia  
[https://en.wikipedia.org/wiki/Three-body\\_problem#Circular\\_restricted\\_three-body\\_problem](https://en.wikipedia.org/wiki/Three-body_problem#Circular_restricted_three-body_problem)
- Gravity assist (the slingshot maneuver) — Wikipedia — how a flyby trades a bit of the  
planet/Moon's orbital energy for spacecraft speed — the physics behind the lunar-  
scatter demo  
[https://en.wikipedia.org/wiki/Gravity\\_assist](https://en.wikipedia.org/wiki/Gravity_assist)
- Specific orbital energy (bound vs. unbound) — Wikipedia — the  $\varepsilon = v^2/2 - \mu/r$  test  
the tool uses to flag escaping objects  
[https://en.wikipedia.org/wiki/Specific\\_orbital\\_energy](https://en.wikipedia.org/wiki/Specific_orbital_energy)
- Hyperbolic trajectory (escape orbits) — Wikipedia  
[https://en.wikipedia.org/wiki/Hyperbolic\\_trajectory](https://en.wikipedia.org/wiki/Hyperbolic_trajectory)
- Butterfly effect / sensitive dependence on initial conditions — Wikipedia — why two  
near-identical launches end up with completely different fates  
[https://en.wikipedia.org/wiki/Butterfly\\_effect](https://en.wikipedia.org/wiki/Butterfly_effect)
- Cislunar space domain awareness (AFRL / SDA overview) — why tracking and keep-  
ing custody of objects out here is a frontier problem  
[https://en.wikipedia.org/wiki/Cislunar\\_space](https://en.wikipedia.org/wiki/Cislunar_space)
- Joseph-Louis Lagrange — Wikipedia  
[https://en.wikipedia.org/wiki/Joseph-Louis\\_Lagrange](https://en.wikipedia.org/wiki/Joseph-Louis_Lagrange)